Tooth-Level Finite-Element Analysis of Serration-Scale Stress Concentration in an *Otodus megalodon* Anterior Tooth

A μCT-to-Mesh Workflow with Bounded Functional Interpretation

Geometry reconstruction, mesh validation, constitutive-model comparison, and explicitly limited palaeobiomechanical inference

DURF Final Technical Report focused on specimen-derived tooth mechanics rather than whole-organism reconstruction

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DURF Final Framework

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Abstract

This DURF final technical report presents a deliberately tooth-level computational workflow from μCT/DICOM voxel data to finite-element analysis of a single anterior tooth assigned to *Otodus megalodon*. The final technical version records the scan voxel spacing used for this report as $(\Delta x, \Delta y, \Delta z) = (50, 50, 100)\ \mu$m; grey values are treated as scanner-native attenuation values rather than medical HU. The manuscript is framed as a definitions-first preliminary technical report: every major claim is explicitly classified as either a direct specimen observation, a model-conditional numerical result, a comparative functional hypothesis, or a deferred organism-scale inference. The direct evidentiary domain is restricted to segmentation, topology repair, TetWild/Gmsh remeshing, Taubin smoothing, curvature-based geometry checks, mesh-quality documentation, and stress-field localisation under stated boundary conditions and material assumptions.
Five constitutive descriptions in the legacy report remain uncalibrated sensitivity models. The original regional Table 9 could not be reproduced because its anatomical masks and 1 MPa stress export were unavailable. A new, independently coded linear-elastic pressure experiment now reports geometry-fixed regional percentiles, archived element IDs, raw fields and numerical checks. Under its explicit 1 MPa face load and bilateral root supports, the two edge-side $Q_{0.95}$ values are 22.373 and 16.809 MPa, versus 11.052 MPa in the central crown reference; the macroscopic apex is 0.540 MPa and the root-side band is 188.756 MPa. Thus this case does not support the former tip-dominant, low-stress-root ordering. These are single-mesh, boundary-condition-dependent results, not evidence of mesh convergence, anatomical serration-tip amplification, fracture, or biological bite pressure. Broader feeding, prey, body-size and ecological interpretations remain outside this single-specimen numerical study.

Project metadata and scope.

Author	Eliguli Han.
Advisor	Leslie Greengard.
Document type	DURF Final Technical Report and tooth-level preliminary manuscript framework; not a finished journal submission.
Primary object	Single anterior tooth of *Otodus megalodon*; tooth-level mechanics are the direct target of the numerical work.
Recorded CT spacing	$(\Delta x, \Delta y, \Delta z) = (50, 50, 100)\ \mu$m for the final DURF report version.
Core pipeline	μCT/DICOM volume; segmentation; marching-cubes surface; topology repair; TetWild/Gmsh volumetric mesh; COMSOL FEA; stress-field and curvature checks.
Main caution	Whole-animal ecology, prey limits, maximum-size estimates, jaw reconstructions, and body-level FEA are excluded from the main inferential chain.

Figure-insertion policy used in this version. Images are grouped as figure plates rather than scattered screenshots. Each plate is placed close to the methodological claim it supports, uses stable scaling relative to the page text width, and has a caption stating what is directly observed versus what is inferred. GUI screenshots are retained only as workflow documentation, while pipeline-rendered tooth-scale figures remain in the main text. Exploratory jaw or whole-body model screenshots are not used as evidentiary results in this version.

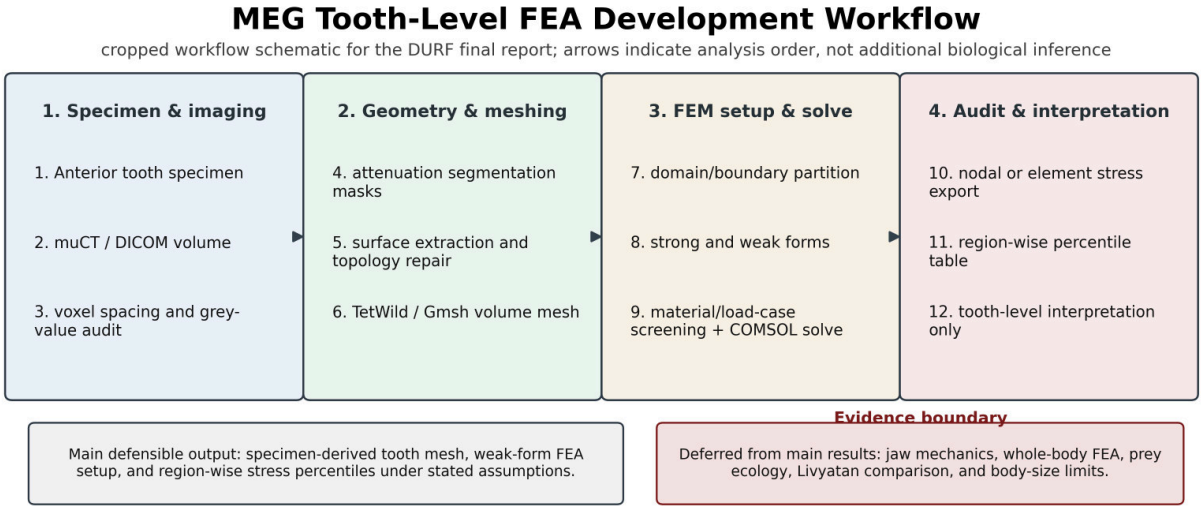


Figure 1: Cropped MEG development workflow inserted as a compact process schematic. The diagram documents the report's internal analysis order: specimen-derived imaging and geometry reconstruction proceed to finite-element weak-form setup, stress-field export, and region-wise percentile audit. The lower evidence-boundary note marks jaw mechanics, whole-body FEA, prey ecology, *Livyatan* comparison, and body-size limits as deferred topics rather than main-result evidence in this tooth-level report.

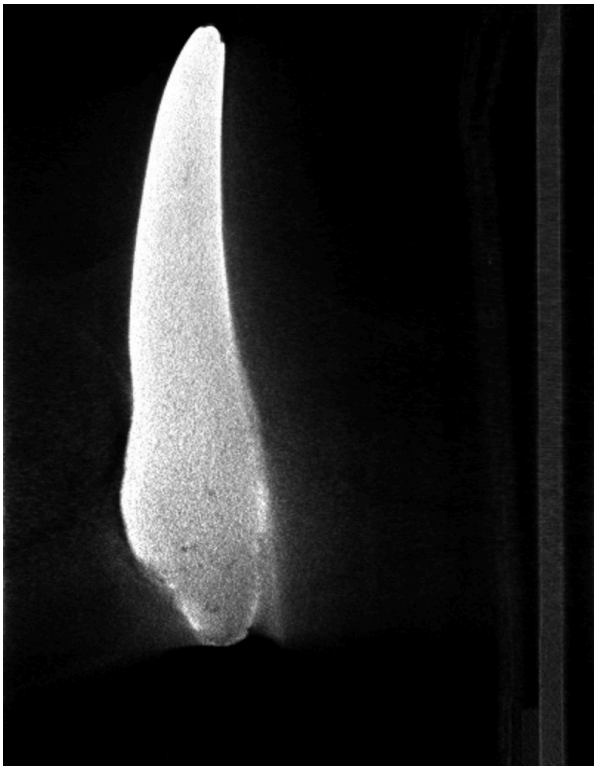


(a) External photograph with user-marked visible edge chip / fracture trace.

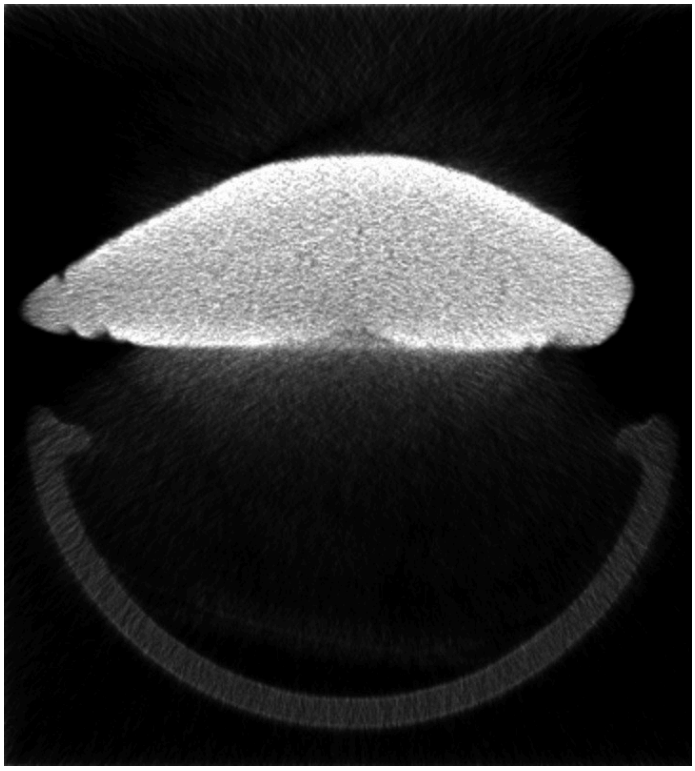


(b) Frontal external photograph of the same single anterior tooth specimen.

Figure 2: Single-specimen photographic documentation. The photographs identify the analysed physical tooth and show visible crown-edge damage relevant to future damage-mapping work. They are used as specimen documentation only; quantitative stress values in this report come from the reconstructed finite-element model, not from photograph-based measurement.



(a) Representative buccolingual slice export from the 0.1 mm slice series.



(b) Representative axial slice export from the 50 μm axial series.

Figure 3: Representative CT slice documentation used to support the recorded anisotropic voxel-spacing entry. For this final DURF report, the working voxel spacing is recorded as $(\Delta x, \Delta y, \Delta z) = (50, 50, 100)\,\mu$m. These slice images document scan geometry and specimen orientation; they do not replace raw DICOM metadata, which should be retained for any external manuscript submission.

Research Summary

This DURF final report is framed as a *tooth-level finite-element case study*, not as a complete reconstruction of *O. megalodon* ecology. Its strongest and most defensible component is the geometry-to-mechanics pipeline: voxel data are segmented, converted to a watertight surface, remeshed, smoothed, checked for curvature distortion, and then analysed under several constitutive descriptions. The mechanically defensible question is narrow: *given this reconstructed anterior tooth, these material parameters, and these loading protocols, where do stress concentrations appear, and how stable are those locations across model and mesh choices?*

All dentition-level, jaw-level, body-level, and ecological interpretations have been removed from the Results structure. They are not treated as outputs of the present FEA. At most, the tooth-level results motivate future comparative tests involving additional tooth positions, damaged fossil specimens, extant lamniform histology, and dynamic tooth-prey contact models. This revision follows the standard palaeobiological evidentiary separation between scanned fossil material and highly reconstructed organism-scale models.

Concrete object record. The computational specimen is treated as a single-tooth solid-mechanics object. The directly auditable quantities are therefore: (i) reconstructed surface topology, (ii) tetrahedral mesh quality, (iii) local curvature preservation after smoothing, (iv) stress-field localisation under the specified loads, and (v) cross-model qualitative stability of the same localisation pattern. The following are deliberately excluded from the main claim set: complete jaw mechanics, whole-body FEA, maximum body-size reconstruction, prey ceiling, thermal niche, competition with *Livyatan*, and Miocene productivity synthesis.

Methodological Scope, Definitions, and Evidentiary Status

The purpose of this revision is not to make stronger claims, but to define more precisely what counts as a valid claim. The manuscript uses the following operational definitions.

Definition 0.1 (Tooth-level FEA claim)**.** A tooth-level FEA claim is a statement about the reconstructed tooth domain Ω_T, its mesh $\mathcal{T}_h$, imposed boundary conditions $\mathcal{B}$, applied load $\mathcal{L}$, constitutive law W, and derived field variables such as displacement $\mathbf{u}$ or von Mises stress σ_{vM}. It is valid only in the form

$$C = C(\Omega_T, \mathcal{T}_h, \mathcal{B}, \mathcal{L}, W),$$

not as a direct statement about whole-animal behaviour or palaeoecology.

Definition 0.2 (Specimen-derived model)**.** A model component is *specimen-derived* if its geometry is obtained from the scanned tooth volume or from deterministic processing of that volume: thresholding, surface extraction, topology repair, smoothing, remeshing, and mesh-quality computation. A component is *comparative* or *reconstructed* if it is imported from an extant analogue, idealised morphology, body outline, jaw reconstruction, or literature-based allometry. Only specimen-derived components are used as primary evidence in this draft.

Definition 0.3 (Model-conditional stress localisation)**.** Let $R \subset \Omega_T$ be a candidate anatomical region, such as a serration-bearing edge band or crown-root transition. A stress-localisation claim means that a summary statistic

$$S_R = \text{Quantile}_q\{\sigma_{\text{vM}}(x) : x \in R\}, \qquad q \in [0.90, 0.99],$$

is elevated relative to an internal reference region R_0 under the stated model choices. This is stronger than visual inspection alone, but weaker than a demonstrated biological failure pathway unless validated against fracture metrics and fossil damage distributions.

Definition 0.4 (Regularised fracture metric)**.** Because sharp or near-sharp serration notches can make pointwise peak stress mesh-dependent, a defensible fracture metric must be regularised. Examples include a percentile stress over a finite process zone $B_R(x_i)$, an annular fit to the leading Williams singular term, an energy-release-rate estimate, or a cohesive-zone damage integral:

$$\overline{\sigma}_R(x_i) = \frac{1}{|B_R(x_i)|} \int_{B_R(x_i)} \sigma_{\text{vM}}(x)\, dx.$$

The present draft reports preliminary localisation and screening indices; it does not yet report a validated convergence study for these regularised fracture metrics.

Definition 0.5 (Evidence levels)**.** Claims are classified into four evidence levels.

Level 1: Direct observation.
Specimen measurements, DICOM-derived geometry, mesh statistics, curvature checks, and solver output generated by the stated pipeline.

Level 2: Model-conditional result.
Quantities that follow only after choosing material parameters, boundary conditions, load magnitude, contact simplification, and mesh regularisation.

Level 3: Comparative functional hypothesis.
Interpretations that require extant lamniform comparison, dental histology, tooth-position homology, tooth-replacement data, or fossil damage surveys.

Level 4: Deferred organism-scale inference.
Statements about prey preference, thermal range, competition, extinction dynamics, complete jaw mechanics, or complete body mechanics. These are outside the evidentiary scope of the present manuscript and are not used as results.

Definition 0.6 (Valid wording rule)**.** A result may be phrased as "shows" or "reports" only for Level 1 output. Level 2 results should be phrased as "under this model", "conditional on", or "in the present simulations". Level 3 interpretations should use "is compatible with", "may have functioned as", or "requires comparative validation". Level 4 claims should be identified as excluded, deferred, or outside the manuscript's scope.

Key revision principle. The text now treats identical high-resolution peak stresses, unextracted notch stress-intensity factors, and uncalibrated hyperelastic parameters as items requiring audit rather than as strong evidence. The acceptable central claim is limited to model-conditional localisation of elevated stress near serrated tooth regions.

Definition 0.7 (Single-specimen finite-element case study)**.** A single-specimen finite-element case study is a computational analysis in which the geometry, mesh, boundary conditions, and stress summaries are reported for one physically available fossil specimen. Such a study can support specimen-specific and model-conditional statements, but it cannot by itself establish species-wide damage frequencies, tooth-position effects, feeding ecology, prey preference, or population-level functional trends. In the present report, the only direct fossil object is one anterior tooth assigned to *Otodus megalodon*. No damaged-megatooth survey, microwear dataset, tooth-position distribution, extant lamniform calibration sample, or population-level ecological dataset is claimed.

Contents

Research Summary 3

Methodological Scope, Definitions, and Evidentiary Status 3

Table 1: Evidence-level audit used in the revised draft. The main text separates specimen-derived tooth mechanics from deferred organism-scale hypotheses and identifies what would be needed to strengthen each claim.

Claim class	Current defensible status	Required before stronger claim
Voxel-to-mesh reconstruction	Level 1: watertightness, mesh statistics, curvature preservation, and qualitative geometry fidelity of the tooth mesh	Independent segmentation repeatability, threshold sensitivity, and raw CT metadata audit
FEA stress field	Level 2: conditional stress localisation under stated loads and materials	Boundary-condition sensitivity, load-area sensitivity, and region-wise percentile stress statistics
Constitutive comparison	Level 2 screening only; current hyperelastic family is tangent-matched rather than independently calibrated	Justification of linear elastic versus nonlinear formulations, calibration target, and material uncertainty propagation
Serration fracture mechanics	Level 2–3 hypothesis: notch geometry is compatible with elevated local stress and possible edge damage	Regularised fracture-metric convergence, EDJ properties, process-zone sensitivity, and damaged fossil comparison
Replacement-mediated function	Level 3 comparative hypothesis, not a result of the FEA alone	Extant lamniform comparison, tooth-position data, tooth-loss mechanics, and fossil damage distributions
Whole organism or palaeoecology	Level 4 deferred; not part of the claim set	Separate jaw/body reconstruction protocol, anatomical priors, uncertainty propagation, and independent palaeoecological evidence

Part I: 3D Reconstruction and Mesh Pipeline 7

1 From Voxel Data to Watertight Surface: Reconstruction Methodology 7
1.1 The Fundamental Problem: Voxels Are Not Geometry . . . 7
1.2 DICOM Acquisition and Segmentation . . . 7
1.3 Watertight Topology: TetWild and Differential-Geometric Compactness . . . 8
1.4 Taubin Smoothing and Curvature Preservation . . . 8
1.5 Curvature Difference Analysis: Tooth-Scale Geometry Validation . . . 8
1.6 Compressed vs. Full-Resolution Meshes . . . 10

Part II: Finite Element Analysis 10

2 Finite-Element Boundary-Value Problem and Weak Form 10

3 Constitutive Model Comparison 11
3.1 Constitutive Sensitivity Family . . . 11
3.2 Material Parameters . . . 12

4 Boundary Conditions, Load Normalisation, and Model Limits 12

5 COMSOL Implementation and Results 12
5.1 Low-Resolution Mesh (~30 000 DOF) . . . 12
5.2 High-Resolution Mesh and Comprehensive Solve Summary (~343 000–2,207,743 DOF) . . . 12
5.3 Exploratory Jaw and Whole-Body Models Excluded from the Main Results . . . 14
5.4 Convergence and the Choice of a Refinement-Stable Target . . . 14

6 Edge-Stress Screening Under Standardised Loading . . . 15
6.1 Geometry-Fixed Region Masks . . . 15
6.2 Region-Wise Percentile and Stress-Amplification Factor . . . 15
6.3 Screening Results Under $\sigma_0 = 1$ MPa . . . 16
6.4 From Screening Index to Fracture Sensitivity . . . 16

7 Discussion: Tooth-Level Functional Interpretation . . . 17
7.1 What the Tooth FEA Can and Cannot Establish . . . 17

7.2 A Limited Replacement-Mediated Cutting Hypothesis . . . 18
7.3 Terminology and Non-Teleological Interpretation . . . 18
7.4 Deferred Inferences . . . 18
7.5 Single-Specimen Evidentiary Status . . . 19
7.6 Deferred Comparative-Ecological Validation Framework . . . 19
7.7 Minimal Manuscript Upgrade Roadmap . . . 21
7.7.1 Route A: biomechanics-first manuscript . . . 21
7.7.2 Route B: palaeobiology / functional-morphology manuscript . . . 21
7.7.3 Seven review-grade modules . . . 21
7.7.4 Five-week minimum work plan . . . 22
7.8 Falsifiable Predictions for Future Work . . . 22

8 Conclusions **22**

Limitations and Future Work **23**

Author Contributions and Supervision **24**

References **24**

A GUI Screenshots and Solver Session Records **24**

Part I: 3D Reconstruction and Mesh Pipeline

1 From Voxel Data to Watertight Surface: Reconstruction Methodology

1.1 The Fundamental Problem: Voxels Are Not Geometry

A μCT scan produces a 3D array of scalar attenuation or grey-value measurements on a regular grid — *voxels*. Unless the scanner output has been explicitly calibrated to the medical CT Hounsfield scale, these values should not be described as Hounsfield units. They are treated here as scanner-dependent attenuation grey values or calibrated attenuation units. This is not geometry in the mechanical sense: finite element analysis requires a boundary-conforming mesh of a continuous solid domain, not a discrete scalar field. The reconstruction problem is therefore a chain of conversions, each introducing approximations that must be controlled:

$$\underbrace{\text{DICOM voxels}}_{\text{scalar field on } \mathbb{Z}^3} \xrightarrow{\text{threshold}} \underbrace{\text{binary mask}}_{\text{occupancy on } \mathbb{Z}^3} \xrightarrow{\text{marching cubes}} \underbrace{\text{triangle soup}}_{\text{non-manifold } \triangle} \xrightarrow{\text{topology repair}} \underbrace{\text{watertight surface}}_{\text{2-manifold } \mathcal{M}} \xrightarrow{\text{TetWild}} \underbrace{\text{tet mesh}}_{\text{FEA-ready } \mathcal{T}} \tag{1}$$

Each arrow is a potential source of geometric error. The curvature difference analysis (fig. 6) quantifies the cumulative error of the entire chain.

1.2 DICOM Acquisition and Segmentation

The source geometry is a μCT scan of an *O. megalodon* anterior tooth (A2 position), segmented from DICOM volumetric data. Earlier drafts used a nominal isotropic label, but the final DURF report records the working voxel spacing as anisotropic:

$$(\Delta x, \Delta y, \Delta z) = (50, 50, 100)\ \mu\text{m}.$$

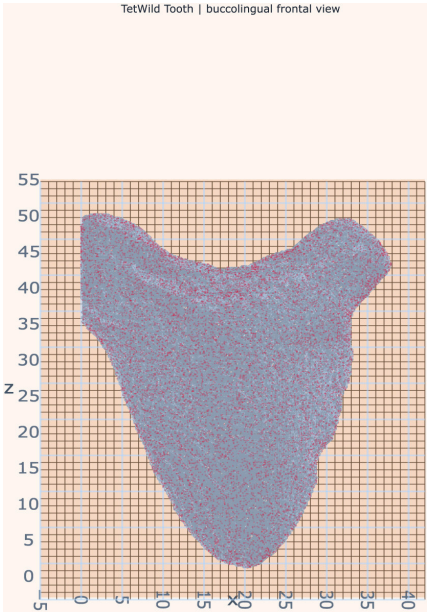
The in-plane spacing is therefore treated as $50\,\mu$m and the through-slice spacing as $100\,\mu$m. For a journal submission, this entry should be accompanied by the raw DICOM spacing tags, scanner log, or reconstruction metadata as the audit trail. The report continues to avoid calling the grey values Hounsfield units unless a scanner calibration record shows that the grey-value scale was calibrated to HU.

Table 2: CT metadata record for the final DURF version. The voxel spacing is now populated as the working scan spacing for this report; raw DICOM tags or scanner logs should still be retained as the external audit trail for manuscript submission.

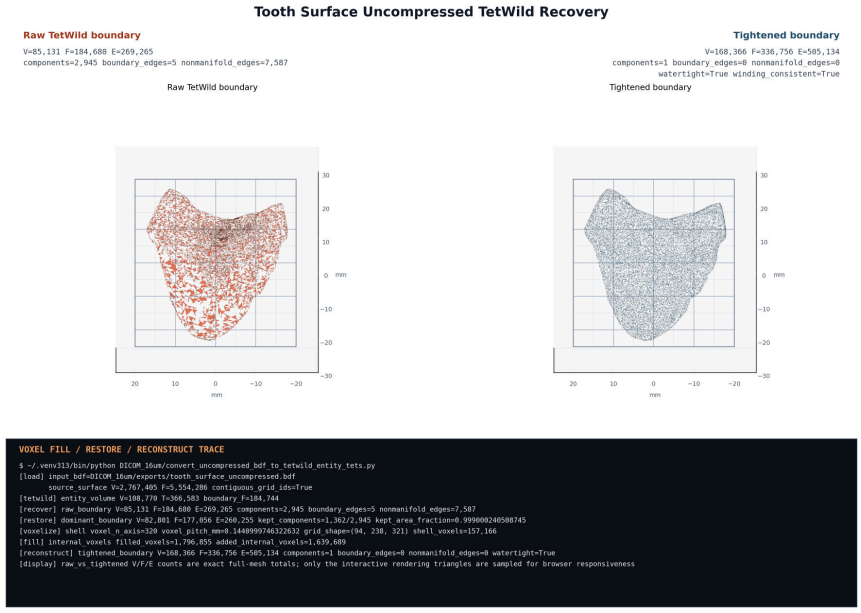
Metadata item	Current status in this draft	Required verification action
Voxel spacing	Filled for this final report as $(\Delta x, \Delta y, \Delta z) = (50, 50, 100)\,\mu$m	Retain DICOM spacing tags, scanner log, reconstruction report, or voxel calibration metadata as audit trail
Verified spacing field	$\Delta x = 50\,\mu$m, $\Delta y = 50\,\mu$m, $\Delta z = 100\,\mu$m	Use the same spacing in segmentation, meshing, exported coordinates, and scale bars
Voxel isotropy	Anisotropic spacing recorded; in-plane spacing differs from through-slice spacing	Avoid isotropic-resolution language; report all three components wherever resolution affects geometry or mesh scale
Grey-value calibration	Treated as relative attenuation / grey value, not HU	Verify whether the scanner output was calibrated to HU or to another attenuation scale
Threshold values	Treated as segmentation parameters, not universal tissue constants	Report threshold units, histogram peaks, and sensitivity to threshold perturbation

The segmentation proceeds in three steps:

1. **Attenuation grey-value thresholding**: the dental μCT histogram is separated into high-attenuation mineralised tooth tissue and background. When tissue subregions are distinguished, threshold values should be reported in the scanner's native grey-value or calibrated attenuation scale. The earlier notation "HU" is therefore removed unless calibration to Hounsfield units is independently documented. For the single-material FEA presented here, mineralised dental tissues are merged into a single solid domain; any enameloid– dentine interface is retained only as a candidate feature for future bi-material or cohesive-zone analysis.
2. **Marching cubes isosurface extraction**: the Lorensen–Cline (1987) marching cubes algorithm extracts a triangle mesh at the enameloid–air isovalue, producing a raw mesh with $\sim 10^6$ triangular faces. This mesh is *not* guaranteed to be manifold: it may contain non-manifold edges (shared by > 2 faces), degenerate triangles (zero area), self-intersections, and topological holes from voxel-level noise.
3. **Topology repair**: non-manifold vertices are split, degenerate faces removed, holes filled by constrained Delaunay triangulation of the boundary loops, and self-intersections resolved by local remeshing. After repair, the surface is a connected, orientable 2-manifold — but still exhibits staircase artefacts from voxel discretisation.



(a) TetWild tooth, buccolingual frontal view. Crown height ∼50 mm, width ∼35 mm. Bilobate root and asymmetric cusplet geometry visible. Grid supports DICOM coordinate registration.



(b) TetWild boundary recovery: raw ($V = 85{,}131$, 2,945 components, 7,587 nonmanifold edges) → tightened ($V = 168{,}366$, 1 component, 0 nonmanifold, watertight=True). Voxel fill trace: 1,796,855 shell + 1,639,689 internal voxels.

Figure 4: Level 1 Voxel-to-mesh reconstruction: (a) smoothed TetWild output with spatial dimensions; (b) boundary topology recovery from non-manifold raw to watertight tightened mesh.

1.3 Watertight Topology: TetWild and Differential-Geometric Compactness

The raw marching-cubes mesh is not guaranteed to be watertight (closed, orientable, manifold). This is fatal for FEA: a non-watertight surface cannot bound a volumetric tetrahedral mesh. We enforce watertightness via TetWild (Hu *et al.*, 2018):

Definition 1.1 (Watertight manifold)**.** A triangle mesh $\mathcal{M} = (V, F)$ is *watertight* if:

1. Every edge is shared by exactly two faces (manifold condition).
2. The mesh is connected and orientable (consistent normal direction).
3. The enclosed volume is well-defined: $\mathrm{Vol}(\mathcal{M}) = \frac{1}{6}\sum_{f\in F} \mathbf{v}_1 \cdot (\mathbf{v}_2 \times \mathbf{v}_3) > 0$.

TetWild produces a robust tetrahedral mesh by:

1. Inserting the input triangles into a background grid.
2. Performing rational-arithmetic-based mesh improvement.
3. Extracting the watertight envelope as a by-product of tetrahedralisation.

1.4 Taubin Smoothing and Curvature Preservation

Raw marching-cubes meshes exhibit staircase artefacts from voxel discretisation. Taubin smoothing (Taubin, 1995) applies alternating positive/negative Laplacian smoothing steps that suppress high-frequency noise without the volume shrinkage of standard Laplacian smoothing:

$$\mathbf{v}_i^{(k+1)} = \mathbf{v}_i^{(k)} + \lambda\, \mathbf{L}(\mathbf{v}_i^{(k)}) \qquad \text{(expansion step)}, \tag{2}$$

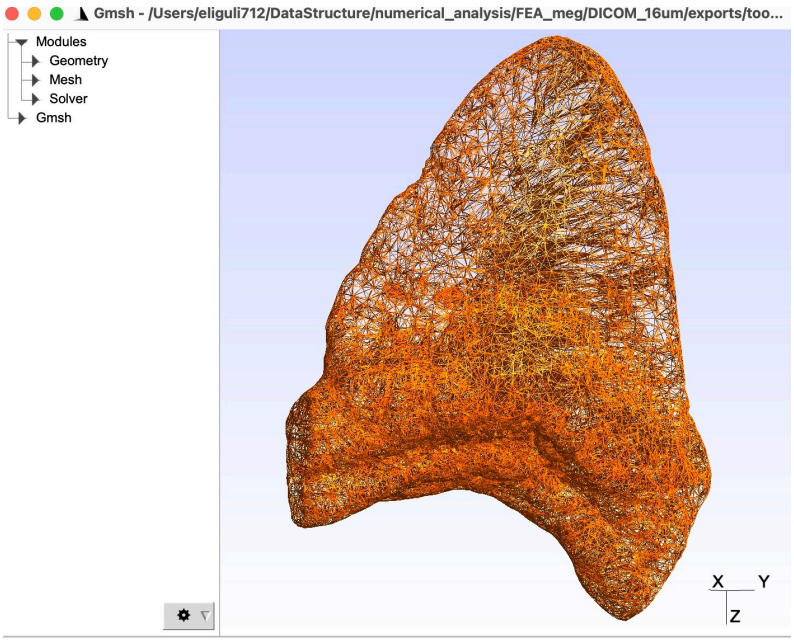
$$\mathbf{v}_i^{(k+2)} = \mathbf{v}_i^{(k+1)} + \mu\, \mathbf{L}(\mathbf{v}_i^{(k+1)}) \qquad \text{(contraction step)}, \tag{3}$$

where $\mathbf{L}$ is the umbrella Laplacian operator, $\lambda > 0$, $\mu < -\lambda < 0$, and the passband is controlled by $k_{\mathrm{PB}} = 1/\lambda + 1/\mu$.

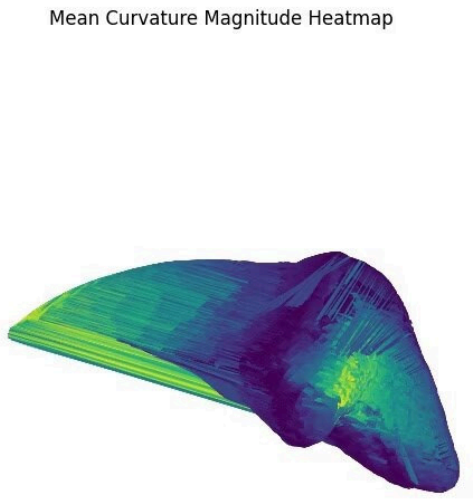
1.5 Curvature Difference Analysis: Tooth-Scale Geometry Validation

The critical validation step is the curvature difference analysis between the compressed (simplified) and original (full-resolution) tooth meshes. This comparison quantifies how much geometric information is lost during mesh decimation before FEA. No organism-scale geometry is used as evidence in the main analysis.

The discrete Gaussian curvature K and mean curvature H are computed at each vertex via the discrete Laplace–Beltrami



(a) Gmsh wireframe overlay on the TetWild-smoothed mesh. Element density is finest near geometrically high-curvature serration and crown-margin regions, which later coincide with elevated stress localisation; the mesh is not claimed to be posteriorly adapted to $\nabla\sigma_{\mathrm{vM}}$.



(b) Mean curvature magnitude $|H|$ heatmap. Peak $|H|$ (yellow) at the serration cutting edge and crown–root junction; bulk crown surfaces near-zero (purple).

Figure 5: Level 1–2 Mesh quality and curvature: (a) pre-solve mesh density concentrated near high-curvature crown-margin and serration regions; (b) mean curvature documenting where geometric detail is retained. Stress localisation is evaluated only after the FEA solve.

operator:

$$H_i = \frac{1}{2}\,\|\Delta_S\,\mathbf{x}_i\|, \qquad K_i = 2\pi - \sum_j \theta_{ij}, \tag{4}$$

where Δ_S is the cotangent Laplacian and θ_{ij} are the angles of the triangles incident on vertex i. The curvature difference $\Delta\kappa = \kappa_{\mathrm{compressed}} - \kappa_{\mathrm{original}}$ is then mapped onto the tooth surface as a scalar field.

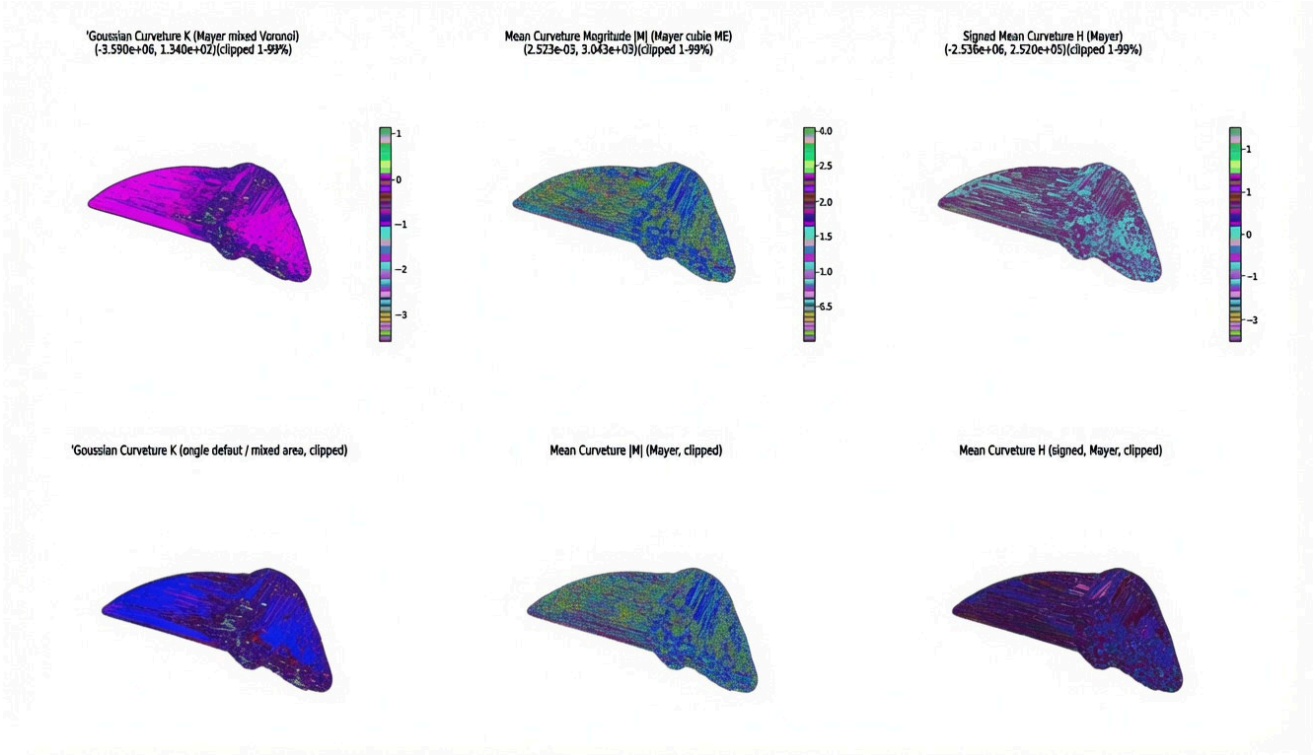


Figure 6: Level 1-2 Six-panel tooth-scale curvature validation. Columns show Gaussian curvature K, unsigned mean-curvature magnitude, and signed mean curvature H; the displayed fields use Meyer mixed-Voronoi/cubic estimators and 1st-99th percentile clipping. Serration ridges retain the strongest curvature extrema, while broad crown surfaces remain comparatively low-curvature.

1.6 Compressed vs. Full-Resolution Meshes

Two mesh resolutions are maintained throughout the analysis:

Table 3: Mesh specifications for the two resolution tiers.

Property	Compressed	Full resolution
Surface triangles	∼8 000	∼120 000
Volume tetrahedra	∼30 000	∼343 000
DOFs (solid mechanics)	∼30 728	∼343 232
Min. element quality	0.12	0.08
Solution time (per study)	∼4 s	∼9 s
Serration resolution	∼3 elements/notch	∼12 elements/notch

Part II: Finite Element Analysis

2 Finite-Element Boundary-Value Problem and Weak Form

This section states the actual mathematical problem being approximated by the finite-element solve. Its purpose is to separate the numerical boundary-value problem from later biological interpretation.

Definition 2.1 (Tooth domain and boundary partition)**.** Let $\Omega_T \subset \mathbb{R}^3$ be the reconstructed solid tooth domain after segmentation, surface repair, and tetrahedralisation. Its boundary is partitioned, up to measure-zero overlaps, as

$$\partial\Omega_T = \overline{\Gamma_D \cup \Gamma_N \cup \Gamma_P}, \qquad \Gamma_i \cap \Gamma_j = \varnothing \quad (i \neq j),$$

where Γ_D is the constrained root or attachment region, Γ_N carries any prescribed traction $\mathbf{t}$, and Γ_P receives an idealised pressure load. The outward unit normal is $\mathbf{n}$. This partition is a modelling choice, not a directly preserved biological boundary condition.

Definition 2.2 (Strong form: finite-deformation static elasticity)**.** For a hyperelastic screening model with stored-energy density $W(\mathbf{F})$, unknown displacement $\mathbf{u} : \Omega_T \to \mathbb{R}^3$, deformation map $\boldsymbol{\varphi}(\mathbf{X}) = \mathbf{X} + \mathbf{u}(\mathbf{X})$, and $\mathbf{F} = \nabla\boldsymbol{\varphi} = \mathbf{I} + \nabla\mathbf{u}$, the first Piola–Kirchhoff stress is

$$\mathbf{P}(\mathbf{F}) = \frac{\partial W}{\partial \mathbf{F}}.$$

The quasi-static boundary-value problem is

$$-\operatorname{Div}\mathbf{P}(\mathbf{F}) = \mathbf{b} \qquad \text{in } \Omega_T, \tag{5}$$

$$\mathbf{u} = \mathbf{u}_D \qquad \text{on } \Gamma_D, \tag{6}$$

$$\mathbf{P}\mathbf{n} = \mathbf{t} \qquad \text{on } \Gamma_N, \tag{7}$$

$$\mathbf{P}\mathbf{n} = -p\mathbf{n} \qquad \text{on } \Gamma_P. \tag{8}$$

Here $\mathbf{b}$ is a body force density, usually set to zero in the present screening simulations, and p is the applied normal pressure magnitude.

Definition 2.3 (Small-strain linear-elastic special case)**.** When displacements and strains are small, the problem reduces to linear elasticity. Define

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}\left(\nabla\mathbf{u} + \nabla\mathbf{u}^T\right), \qquad \boldsymbol{\sigma}(\mathbf{u}) = \mathbb{C} : \boldsymbol{\varepsilon}(\mathbf{u}),$$

where $\mathbb{C}$ is the fourth-order elasticity tensor determined by E and ν, or equivalently by Lamé parameters λ and μ. The strong form is

$$-\nabla \cdot \boldsymbol{\sigma}(\mathbf{u}) = \mathbf{b} \qquad \text{in } \Omega_T, \tag{9}$$

$$\mathbf{u} = \mathbf{u}_D \qquad \text{on } \Gamma_D, \tag{10}$$

$$\boldsymbol{\sigma}(\mathbf{u})\mathbf{n} = \mathbf{t} \qquad \text{on } \Gamma_N, \tag{11}$$

$$\boldsymbol{\sigma}(\mathbf{u})\mathbf{n} = -p\mathbf{n} \qquad \text{on } \Gamma_P. \tag{12}$$

Definition 2.4 (Weak form of the tooth-level elasticity problem)**.** Let

$$V = \{\mathbf{v} \in H^1(\Omega_T)^3 : \mathbf{v} = \mathbf{u}_D \text{ on } \Gamma_D\}, \qquad V_0 = \{\mathbf{v} \in H^1(\Omega_T)^3 : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_D\}.$$

The finite-deformation weak problem is: find $\mathbf{u} \in V$ such that

$$\int_{\Omega_T} \mathbf{P}(\mathbf{I} + \nabla\mathbf{u}) : \nabla\mathbf{v}\, dV = \int_{\Omega_T} \mathbf{b}\cdot\mathbf{v}\, dV + \int_{\Gamma_N} \mathbf{t}\cdot\mathbf{v}\, dS - \int_{\Gamma_P} p\,\mathbf{n}\cdot\mathbf{v}\, dS \qquad \forall\,\mathbf{v} \in V_0. \tag{13}$$

For the linear-elastic case, $\mathbf{P}$ is replaced by the Cauchy stress in the bilinear form:

$$\int_{\Omega_T} \boldsymbol{\sigma}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v})\, dV = \int_{\Omega_T} \mathbf{b}\cdot\mathbf{v}\, dV + \int_{\Gamma_N} \mathbf{t}\cdot\mathbf{v}\, dS - \int_{\Gamma_P} p\,\mathbf{n}\cdot\mathbf{v}\, dS \qquad \forall\,\mathbf{v} \in V_0. \tag{14}$$

Equations (13)–(14) are the mathematical objects approximated by the FEA; any biological interpretation is downstream of these choices.

Definition 2.5 (Finite-element discretisation)**.** Let $\mathcal{T}_h$ be a conforming tetrahedral mesh of Ω_T with characteristic element size h. For polynomial degree r (typically $r = 1$ or 2 in standard solid-mechanics tetrahedral elements), define

$$V_h = \{\mathbf{v}_h \in V : \mathbf{v}_h|_K \in \mathbb{P}_r(K)^3 \text{ for every } K \in \mathcal{T}_h\}, \qquad V_{h,0} = V_h \cap V_0.$$

The discrete problem is: find $\mathbf{u}_h \in V_h$ such that the residual

$$R_h(\mathbf{u}_h; \mathbf{v}_h) = 0 \qquad \forall\,\mathbf{v}_h \in V_{h,0},$$

with R_h given by the left-minus-right side of (13) or (14). Nonlinear material models require Newton or quasi-Newton iteration; linear elasticity requires one linear solve after assembly.

Definition 2.6 (Reported stress output)**.** The von Mises field used for visualisation is a post-processed scalar computed from the stress tensor,

$$\sigma_{\mathrm{vM}} = \sqrt{\frac{3}{2}\,\mathbf{s} : \mathbf{s}}, \qquad \mathbf{s} = \boldsymbol{\sigma} - \frac{1}{3}\,\mathrm{tr}(\boldsymbol{\sigma})\mathbf{I}.$$

It is a useful scalar summary of distortional stress, but it is not by itself a fracture criterion for dental tissues. Fracture interpretation requires additional assumptions about flaws, toughness, interfaces, or a cohesive-zone law.

3 Constitutive Model Comparison

3.1 Constitutive Sensitivity Family

The tooth is analysed under one linear elastic formulation and several nonlinear hyperelastic descriptions. In this draft, these models are not presented as independently calibrated biological material laws for fossil enameloid. They are a numerical sensitivity family used to test whether the *spatial pattern* of stress localisation persists when the tangent stiffness is held approximately comparable.

Definition 3.1 (Tangent-matched constitutive sensitivity family)**.** A constitutive family $\{W_k\}_{k=1}^m$ is tangent-matched at the reference state if each model has approximately the same small-strain modulus and Poisson ratio at $\mathbf{F} = \mathbf{I}$. Such a family is useful for checking stress-field robustness in small-strain simulations, but it does not demonstrate that the material is truly hyperelastic or that different large-strain laws have been biologically validated.

The five model forms used for this numerical check are:

Definition 3.2 (Constitutive model suite)**.** Let $\mathbf{F}$ be the deformation gradient, $\mathbf{C} = \mathbf{F}^T\mathbf{F}$ the right Cauchy–Green tensor, $J = \det\mathbf{F}$, and $\bar{I}_1$, $\bar{I}_2$ the isochoric invariants.

(i) St. Venant–Kirchhoff (StVK).

$$W_{\mathrm{StVK}} = \frac{\lambda}{2}(\mathrm{tr}\,\mathbf{E})^2 + \mu\,\mathrm{tr}(\mathbf{E}^2), \qquad \mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}). \tag{15}$$

(ii) Neo-Hookean.

$$W_{\mathrm{NH}} = \frac{\mu}{2}(\bar{I}_1 - 3) + \frac{\kappa}{2}(J-1)^2. \tag{16}$$

(iii) Ogden ($N = 1$).

$$W_{\mathrm{Og}} = \sum_{p=1}^{N} \frac{\mu_p}{\alpha_p}(\bar{\lambda}_1^{\alpha_p} + \bar{\lambda}_2^{\alpha_p} + \bar{\lambda}_3^{\alpha_p} - 3) + \frac{\kappa}{2}(J-1)^2. \tag{17}$$

(iv) Mooney–Rivlin (MR2, 2-parameter).

$$W_{\mathrm{MR2}} = C_{10}(\bar{I}_1 - 3) + C_{01}(\bar{I}_2 - 3) + \frac{\kappa}{2}(J-1)^2. \tag{18}$$

(v) Mooney–Rivlin (MR5, 5-parameter).

$$W_{\mathrm{MR5}} = \sum_{i+j\le 2} C_{ij}(\bar{I}_1 - 3)^i(\bar{I}_2 - 3)^j + \frac{\kappa}{2}(J-1)^2. \tag{19}$$

3.2 Material Parameters

Table 4: Material parameters assigned to the single-material tooth domain. The nonlinear entries are tangent-matched screening parameters, not independently calibrated fossil-tissue laws. They should be replaced by a material-uncertainty analysis in a submission-ready manuscript.

Model	Parameters	Status
StVK	$\lambda = 29.0\,\mathrm{GPa}$, $\mu = 28.5\,\mathrm{GPa}$	literature-derived elastic target
Neo-Hookean	$\mu = 28.5\,\mathrm{GPa}$, $\kappa = 48.6\,\mathrm{GPa}$	tangent-matched screening value
Ogden	$\mu_1 = 57.0\,\mathrm{GPa}$, $\alpha_1 = 2.0$	tangent-matched screening value
MR2	$C_{10} = 20.0\,\mathrm{GPa}$, $C_{01} = 8.5\,\mathrm{GPa}$	tangent-matched screening value
MR5	5-parameter screening set	tangent-matched screening value

4 Boundary Conditions, Load Normalisation, and Model Limits

The FEA results should be interpreted as comparative stress-localisation experiments rather than direct reconstructions of a single biological bite. The load cases are normalised so that stress patterns, amplification zones, and model-to-model differences can be compared under controlled conditions. Absolute stress magnitudes depend on the assumed load area, constraint placement, material stiffness, and whether the enameloid-dentine interface is treated as continuous or damageable.

Assumption 4.1 (Core FEA assumptions)**.** The present simulations assume: (i) a merged single-material dental solid for the main solve; (ii) literature-derived elastic or hyperelastic parameters; (iii) idealised pressure or displacement loads; (iv) fixed or partly constrained root regions approximating attachment to the jaw; and (v) no explicit prey tissue, saliva, cyclic fatigue, or tooth-row contact. Each assumption is acceptable for a first stress-localisation study, but each limits behavioural interpretation.

Remark 4.1 (Interpreting peak stress). Peak von Mises stress near a serration tip is not expected to be fully mesh-independent because sharp notches introduce local singular behaviour. For this reason, the revised interpretation emphasises the *location and persistence* of high-stress regions and treats regularised fracture metrics, not raw pointwise maxima, as the appropriate target for future convergence analysis.

5 COMSOL Implementation and Results

5.1 Low-Resolution Mesh (∼ 30 000 DOF)

Results from the 30k DOF mesh under all five constitutive models are tabulated below; COMSOL GUI screenshots are in fig. 11.

Table 5: Low-resolution (30k DOF) peak von Mises stress by constitutive model.

Model	$\sigma_{\mathrm{vM}}^{\max}$ (Pa)	Solve time
StVK (std1)	1.6×10^4	4 s
Neo-Hookean	1.2×10^4	4 s
Ogden	1.6×10^4	4 s
MR2	2.0×10^4	4 s
MR5	2.1×10^4	4 s

5.2 High-Resolution Mesh and Comprehensive Solve Summary (∼ 343 000–2,207,743 DOF)

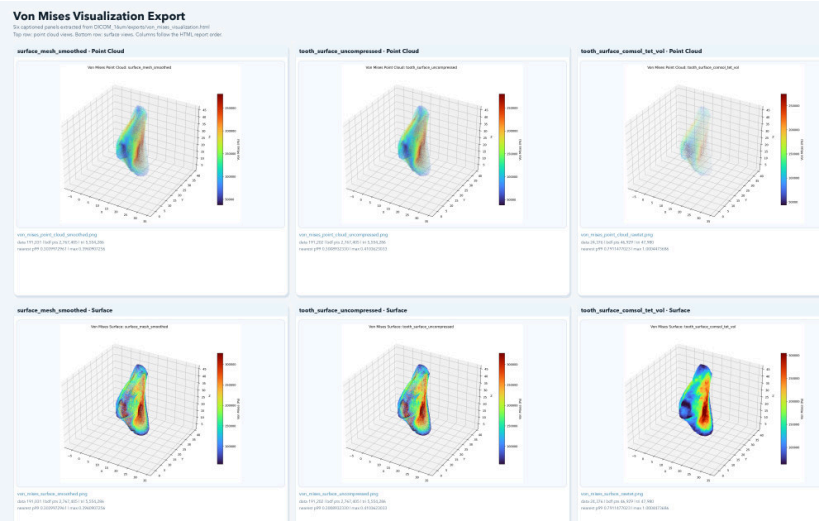
Remark 5.1 (High-resolution peak-stress audit). A table in which all high-resolution peak stresses are numerically identical is a warning sign rather than a strong result. It could reflect small-strain convergence to the same tangent response, but it could also reflect a shared post-processing dataset, copied export value, or insufficiently separated COMSOL studies. This report therefore uses the high-resolution table only for traceability and relies on spatial stress localisation, not exact peak comparison, for the central interpretation.

Table 6: High-resolution post-processing audit for the tooth-level solve. The identical reported peak values across studies are retained as a traceability record, but they are *not* used as evidence that all constitutive laws produced identical mechanical responses. The values should be checked against raw COMSOL datasets before manuscript submission.

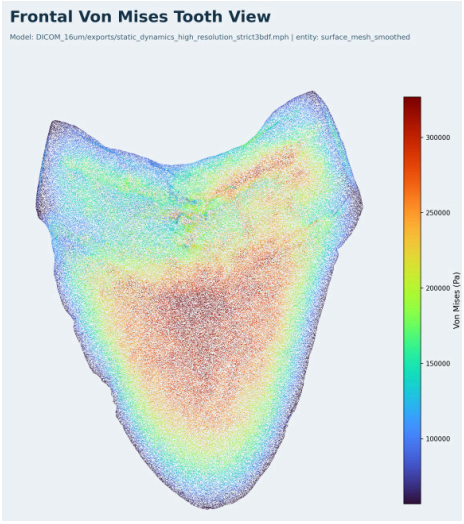
Study	Observed DOF	Mesh tetrahedra	Reported peak σ_{vM} (Pa)	Current interpretation
Linear elastic	2,207,743	1,366,270	2.185×10^6	solver trace; verify dataset identity
Neo-Hookean	2,207,743	1,366,270	2.185×10^6	solver trace; verify nonlinear export
Ogden	2,207,743	1,366,270	2.185×10^6	solver trace; verify parameter set
Mooney–Rivlin MR2	2,207,743	1,366,270	2.185×10^6	solver trace; verify independent study
Mooney–Rivlin MR5	2,207,743	1,366,270	2.185×10^6	solver trace; verify independent study
Pressure MR5	2,207,743	1,366,270	2.185×10^6	solver trace; verify load case

Table 7: Stress-output audit after numerical re-evaluation. Distinct load cases and sampling entities must remain separate.

Reported scale	Source / context	Audited interpretation
1.2–2.1×10^4 Pa	Legacy 30k-DOF studies	Retained for traceability; not compared directly with the fresh pressure case.
2.185×10^6 Pa	Legacy high-resolution summary	Identical study maxima remain unverified; no constitutive-equivalence conclusion.
$\sim$ 300 kPa rendered; 0.253390 MPa $Q_{0.95}$	Archived smoothed export / old frontal view	Whole-export sample percentile, not a regional statistic or a verified 1 MPa pressure result. Exporter source configures an MR5 body load.
18–52 MPa	Former pressure-normalised regional table	Withdrawn as a computational result: original masks, pressure-case field and reproducible command are unavailable.
0.540–188.756 MPa $Q_{0.95}$	New Table 9: P1 linear elasticity, 1 MPa face pressure	Newly defined geometric masks on 55,852 tetrahedra; A/B orientation and individual serration apices unverified.
0.39 MPa	Former LEFM illustration	Arithmetic only; the assumed SAF of 52 was not validated. No critical-pressure result is retained.



(a) Six-panel σ_{vM} export: point cloud (top) and surface (bottom) for smoothed, uncompressed, and COMSOL tet_vol meshes. Nearest p99: 0.30–0.41 (smoothed/uncompressed), 0.79–1.00 (rawtet).



(b) Frontal σ_{vM} on `surface_mesh_smoothed` entity. Peak ∼300 kPa (dark red) at crown centre; serration-edge band green/yellow ∼150–200 kPa; root lobes minimal (blue, <80 kPa).

Figure 7: Pipeline-rendered von Mises stress visualisation.

5.3 Exploratory Jaw and Whole-Body Models Excluded from the Main Results

Earlier internal drafts included exploratory jaw and whole-body COMSOL screenshots. Those models are deliberately excluded from the present Results section. A complete *O. megalodon* body, soft-tissue volume, cranial cartilage distribution, musculature, and caudal peduncle geometry are not directly preserved in the scanned tooth and would require a separate comparative reconstruction protocol. Consequently, any such model is not comparable in evidentiary status to the μCT-derived tooth mesh and is not used here to infer organismal weak points, swimming mechanics, prey interaction, or strike dynamics.

This exclusion is methodological rather than substantive: future work may construct a jaw- or body-level model, but it should be published as a separate comparative reconstruction with explicit anatomical priors, uncertainty propagation, and sensitivity tests.

5.4 Convergence and the Choice of a Refinement-Stable Target

Peak von Mises stress is not, by itself, an adequate convergence target near serration tips. Each notch is a sharp or near-sharp re-entrant corner, and under the linear-elastic idealisation the leading near-tip term of the field is

$$\sigma_{ij}(r,\theta) \sim K_N r^{\lambda-1} \widetilde{\sigma}_{ij}(\theta;\alpha), \qquad \lambda - 1 < 0, \tag{20}$$

so that $\sigma_{\mathrm{vM}} \to \infty$ as $r \to 0$. On a mesh of characteristic size h the sampled peak therefore scales as

$$\sigma_{\max}(h) \sim C h^{-(1-\lambda)} \xrightarrow[h\to 0]{} \infty. \tag{21}$$

That is, the single-node maximum diverges under refinement rather than converging. Reporting $\sigma_{\max}$ as a structural quantity would record a mesh artefact, not a property of the specimen; this behaviour is intrinsic to the sharp-notch idealisation and is not removed by a finer mesh.

The consequence is a criterion for admissibility rather than a numerical inconvenience: any quantity intended to converge under refinement must be a bounded functional of the field over a set of finite measure. Two such targets are used in this project, in ascending order of ambition:

Continuation of Section 5.4. Two possible reporting targets are a regional percentile $Q_q(R_i)$ and an annular fit to the notch-intensity coefficient K_N. A percentile can reduce sensitivity to an isolated extreme sample, but is not automatically bounded uniformly across mesh levels or converged. In particular, adaptive refinement changes an equal-weight node/element empirical distribution. Fixed physical masks, a stated sampling measure and an explicit mesh study are required.

Definition 5.2 (Candidate notch-intensity estimate). An estimate of K_N may be fitted on an annulus outside the singular tip element and inside the notch-dominated field. A convergence claim requires at least three mesh levels and several annuli. These fits are not available here.

Remark 5.3 (Current status). Neither the former 5% K_N agreement nor region-percentile mesh convergence has been demonstrated. The new calculation below establishes finite-mesh equilibrium and precisely reproducible statistics, not convergence.

6 Edge-Stress Screening Under Standardised Loading

The original Table 9 analysis is not reproducible from the available archive: its six anatomical index sets and standardised-pressure stress export are missing. After that audit, a new pressure experiment was designed and executed. It uses the largest face-connected component of the archived refined TetWild volume mesh, homogeneous small-strain P1 elasticity, $\lambda = 29.0$ GPa, $\mu = 28.5$ GPa, and $\sigma_0 = 1$ MPa. Seven detached or hinged tetrahedra were excluded, totalling 0.000300% of input volume; 13,584 nodes and 55,852 tetrahedra remain.

6.1 Geometry-Fixed Region Masks

All masks are determined from geometry and saved before solving. Exact solid-volume PCA defines longitudinal position $s \in [0, 1]$ from the macroscopic apex to the root end; its sign agrees with the scan $+Z$ direction. The local transverse silhouette coordinate r uses 40 longitudinal bins, and ζ is the normalised thickness coordinate. These are proxies, not recovered anatomical annotations.

Table 8: New geometric masks, stored as zero-based tetrahedral index sets.

Mask	Fixed rule
R_0	$0.25 \le s \le 0.55$, $\lvert r\rvert \le 0.35$, $\lvert\zeta\rvert \le 0.5$; central crown reference.
R_A	$0.12 < s < 0.64$, $r \le -0.78$; one edge-side band.
R_B	$0.12 < s < 0.64$, $r \ge 0.78$; the other edge-side band.
R_{tip}	$s \le 0.12$; macroscopic apex only.
R_{cr}	$0.64 \le s \le 0.76$; geometric transition band.
R_{root}	$0.78 \le s \le 0.84$; root-side band below the supports.

Pressure acts on exterior faces with centroid $0.18 \le s \le 0.64$, $\lvert r\rvert \le 0.75$ and outward-normal alignment at least 0.5 with the positive thickness direction; traction is $-\sigma_0\mathbf{n}$. The load area is 418.091 mm^2. All three translations are fixed at the top cap of each root lobe, measured from that lobe's own endpoint; the baseline cap depth is 0.04 times longitudinal length. The loaded face is not independently identified as labial rather than lingual.

6.2 Region-Wise Percentile and Stress-Amplification Factor

Definition 6.1. For this finite-mesh field, let n_i be the number of elements in R_i. The primary statistic uses the equal-weight linear/type-7 empirical quantile of constant element stresses:

$$Q_q(R_i) = \text{Quantile}_q^{\text{linear}}\{\sigma_{\text{vM},e} : e \in R_i\}, \qquad q \in \{0.90, 0.95, 0.99\}. \tag{22}$$

Volume-weighted inverse-CDF quantiles are archived separately to expose sampling-measure sensitivity. The regional masks are not an exhaustive anatomical partition.

Definition 6.2 (Stress-amplification factor). When the nominal applied pressure is verified for the actual solve,

$$\mathrm{SAF}_q(R_i) = \frac{Q_q(R_i)}{\sigma_0}. \tag{23}$$

The denominator is the applied pressure, not the crown-region percentile. Stress values must be extracted from the archived element field, never from a rendered colour map. For a sorted sample $v_0 \le \cdots \le v_{n-1}$, the linear quantile interpolates at $h = (n-1)q$ between $v_{\lfloor h \rfloor}$ and $v_{\lceil h \rceil}$.

6.3 Screening Results Under $\sigma_0 = 1$ MPa

The following table replaces the unsupported original values with a newly executed, explicitly different linear-elastic pressure experiment. It does not recreate the lost original anatomical masks or constitutive study.

Table 9: Recalculated region-wise percentile stress and stress-amplification factors under a standardised crown-face pressure $\sigma_0 = 1$ MPa. Regions are newly defined geometric proxies. Values are equal-weight element summaries of σ_{vM} (MPa), not pointwise fracture criteria.

Region	n_e	$Q_{0.90}$	$Q_{0.95}$	$Q_{0.99}$	$\mathrm{SAF}_{0.95}$
Central crown reference R_0	1,186	9.725	11.052	13.140	11.052
Edge-side A R_A	1,606	20.183	22.373	30.605	22.373
Edge-side B R_B	1,050	13.911	16.809	19.610	16.809
Macroscopic apical tip R_{tip}	723	0.409	0.540	0.779	0.540
Transition proxy R_{cr}	21,752	31.806	39.214	56.280	39.214
Root-side band R_{root}	6,801	134.899	188.756	237.625	188.756

n_e is the number of tetrahedra in the mask. For this verified 1 MPa load, SAF is numerically equal to Q in MPa. A/B anatomical mesial/distal identity and the labial/lingual identity of the loaded face remain unverified. The apical mask is the macroscopic tooth tip, not individual serration-tip process zones.

Interpretation. In this case the ordering of $Q_{0.95}$ is root-side band > transition > side A > side B > central crown > macroscopic apex. The two edge-side-to-crown ratios are 2.024 and 1.521, not 20–50. The apex mask is not a serration-tip process zone. The elevated root-side values reflect this supported bending configuration and cannot identify biological failure sites or establish a universal anatomical ordering.

Numerical validation and sensitivity. The affine-traction patch-test error is 3.96×10^{-15}. The main solve has relative free-DOF residual 5.11×10^{-11}, force-balance error 1.43×10^{-12}, torque-balance error 5.79×10^{-13} and energy-identity error 2.88×10^{-12}. Pressure doubling gives exactly doubled computed displacements in this linear solve. These checks verify the implemented finite system, not biological validity.

With bilateral cap depths 0.03, 0.04 and 0.05, edge-side A $Q_{0.95}$ is 22.373–22.888 MPa, side B 16.116–16.809 MPa, and the root-side band 176.883–190.933 MPa. Changing the edge-mask threshold from $|r| = 0.70$ to 0.86 gives side A 21.756–23.503 MPa and side B 14.237–18.126 MPa. These are bounded sensitivity checks on one mesh, not a convergence study.

Archive audit. The older smoothed field has 191,031 samples, whole-export $Q_{0.95} = 0.253390$ MPa and maximum 0.401620 MPa. Seventeen of the former table's 18 Q values exceed that maximum and cannot be subsets of this unscaled field. The exporter source instead configures an MR5 body load of 5×10^4 N/m^3; no pressure-equivalent rescaling is justified.

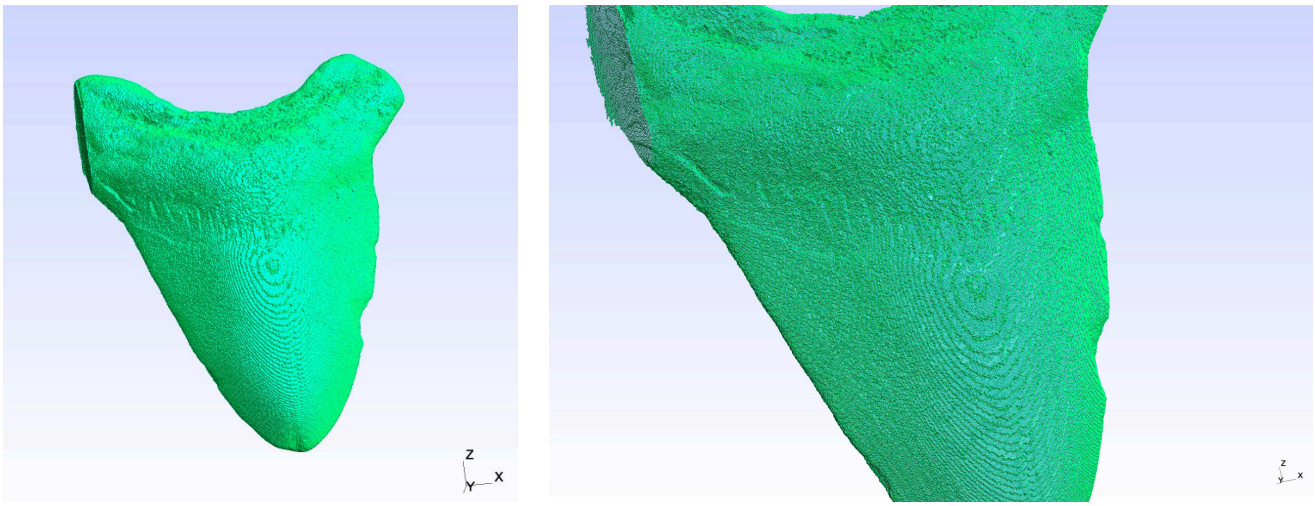
6.4 Limits of Fracture Screening

The formal sensitivity index and nominal-pressure expression may be retained only as assumed LEFM screening relations:

$$\Phi_i(a) = \frac{\mathrm{SAF}_i \sigma_0 \sqrt{\pi a}}{K_{\mathrm{IC}}}, \tag{24}$$

$$\sigma_0^{\mathrm{crit}} = \frac{K_{\mathrm{IC}}}{\mathrm{SAF}_i \sqrt{\pi a}}. \tag{25}$$

No numerical critical pressure or fracture conclusion is inferred from Table 9. The earlier 0.39 MPa illustration depended on an unvalidated SAF of 52. Mesh convergence, anatomical region validation and appropriate crack-driving stress/toughness data remain necessary.



(a) High-magnification view of the TetWild-smoothed crown. The image documents surface geometry and residual reconstruction texture. It does not measure a serration SAF; the previous 40–52 attribution is withdrawn.

(b) Gmsh wireframe overlay documents geometry-driven mesh density. The new Table 9 uses a specified volume mesh and geometric masks. Visual mesh density alone neither establishes stress elevation nor demonstrates convergence.

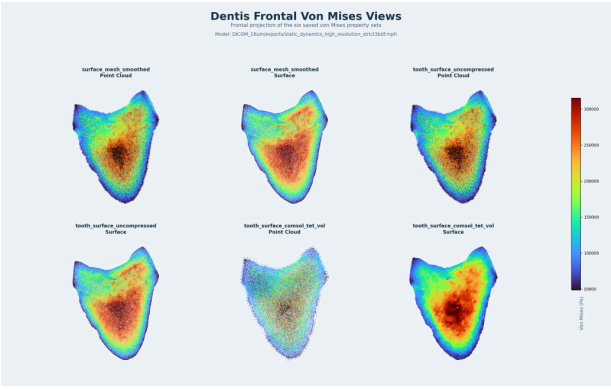
Figure 8: Level 1–2 Serration substructure geometry (a) and geometry-driven mesh-density distribution (b).

7 Discussion: Tooth-Level Functional Interpretation

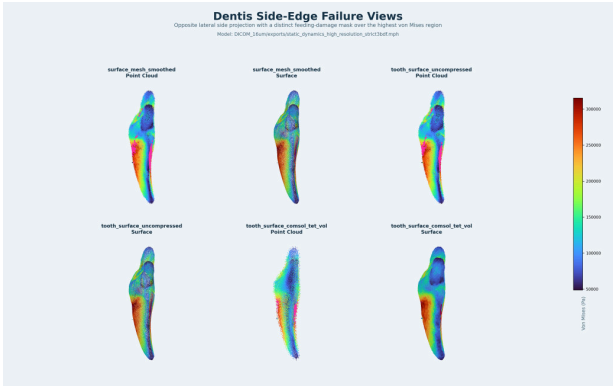
7.1 What the Tooth FEA Can and Cannot Establish

The revised pressure experiment supports only a finite-mesh, load- and support-dependent regional comparison. Edge-side A/B $Q_{0.95}$ exceeds the central crown reference by factors of 2.024/1.521, while the macroscopic apex is lower and the root-side band is higher. The missing original masks prevent verification of the former serration-tip ordering.

Damage-tolerant cutting, chipping and EDJ spalling remain hypotheses requiring anatomical and material validation. Neither a stress percentile nor an archived visualisation establishes crack initiation, a feeding sequence, biological bite pressure or ecological function.



(a) Frontal σ_{vM} views: six property sets (smoothed point cloud/surface, uncompressed point cloud/surface, COMSOL tet_vol point cloud/surface). Peak ∼300 kPa at crown centre; serration edges and EDJ clearly delineated.



(b) Ventral (occlusal) σ_{vM} views: same six property sets. The cutting-edge stress band is visible as a continuous high-σ_{vM} arc tracing the serration line from mesial to distal.

Figure 9: Multi-view von Mises stress atlas across three mesh representations (smoothed, uncompressed, COMSOL volumetric) and two rendering modes (point cloud, surface interpolation). The stress pattern appears consistent across all six variants, supporting a preliminary claim of spatial localisation rather than full fracture-metric convergence.

7.2 A Limited Replacement-Mediated Cutting Hypothesis

Popular descriptions often treat shark tooth replacement as making individual tooth damage irrelevant. The tooth-level FEA suggests a more precise hypothesis: serration-scale stress concentration may have made local edge damage more likely under oblique or high-load contact, while continuous replacement could have reduced the long-term functional cost of that damage. This is a hypothesis about functional morphology, not a proof of a complete predatory strategy.

The lateral bending scale used in the earlier draft is retained only as a geometric comparison:

$$F_{\mathrm{crit}}^{(\mathrm{lateral})} \propto WH\hat{\tau}^2, \tag{26}$$

where W is crown width, H is crown height, and $\hat{\tau}$ is normalised labiolingual thickness. This equation is useful because it separates tooth shape from body-level inference. At constant W and H, a thinner crown has lower lateral bending resistance; at natural sizes, absolute force capacity also depends on the larger tooth dimensions. The result should therefore be read as a morphological scaling argument, not as a direct estimate of prey-killing performance.

7.3 Terminology and Non-Teleological Interpretation

The following definitions fix the vocabulary used in the Discussion.

Definition 7.1 (Damage-tolerant cutting)**.** In this draft, "damage-tolerant cutting" means that localised edge damage or EDJ-associated spalling could remain compatible with tooth function in a dentition that undergoes continuous replacement. It does not mean that tooth damage is beneficial in every case, that tooth loss was intentional, or that a complete predatory sequence has been reconstructed.

Definition 7.2 (Replacement-mediated functional hypothesis)**.** A replacement-mediated functional hypothesis links tooth morphology, localised mechanical loading, and tooth replacement only at the level of a comparative evolutionary hypothesis. It requires external validation from extant lamniform replacement mechanics, fossil damage distributions, and tooth-position comparisons.

To keep the wording aligned with evolutionary functional morphology, this version avoids engineering teleology. Phrases such as "designed failure point", "calibrated to the predation mode", "normal operating condition", and "deliberate design choice" are replaced by cautious terms such as "mechanically consistent with", "compatible with", "may have functioned as", and "requires comparative validation". Natural selection can produce functional compromises, but fossil morphology should not be written as though an engineer intentionally designed it.

7.4 Deferred Inferences

The following topics are excluded from the evidentiary chain of the present paper:

1. complete tooth-row mechanics;
2. jaw-cartilage reconstruction and bite-force estimation;
3. complete-body FEA or caudal-peduncle weak-point claims;
4. complete predatory strike modelling;
5. prey-size ceilings, prey preference, or wound-and-wait behaviour;
6. competition or niche partitioning with *Livyatan*;

7. Miocene productivity, climate, or extinction synthesis.

These topics may be appropriate for a later Discussion paper or supplementary comparative model, but they require independent anatomical and palaeoecological evidence. They are not direct consequences of a scanned single-tooth FEA.

7.5 Single-Specimen Evidentiary Status

The present report is based on a single physically available *Otodus megalodon* anterior tooth specimen. Consequently, all quantitative finite-element results reported here are specimen-specific and should be interpreted as a tooth-level computational case study rather than as a population-level survey of megatooth damage, tooth-position variation, or species-wide feeding ecology.

No fossil damage-frequency dataset, tooth-position distribution dataset, microwear dataset, or comparative extant lamniform calibration dataset is claimed in the present version. Ecological and functional interpretations are therefore restricted to model-conditional compatibility statements: the observed stress-localisation pattern is mechanically compatible with serration-mediated cutting and local chipping hypotheses, but it does not by itself establish feeding behaviour, prey preference, tooth replacement rate, or species-wide fracture frequency.

A manuscript intended for palaeobiological publication would require expansion to a multi-specimen comparative dataset, including damaged megatooth specimens, tooth-position assignments, lamniform histological comparison, enameloid-dentine junction material-parameter constraints, and extant analogue calibration. These missing datasets do not invalidate the present finite-element case study; they delimit the level of inference.

7.6 Deferred Comparative-Ecological Validation Framework

Because the present study contains only one fossil tooth specimen, the ecological component is treated as a future validation framework rather than as a result of the present analysis. A complete palaeobiomechanical manuscript would require at least four additional evidentiary layers:

1. a damaged megatooth specimen survey recording fracture position, chipping morphology, preservation state, and tooth position where assignable;
2. a tooth-position distribution analysis separating anterior, lateral, and posterior teeth, because stress localisation and functional loading cannot be assumed invariant across the dentition;
3. histological and material-property constraints for enameloid, dentine, and the enameloid-dentine junction;
4. extant lamniform analogue calibration using modern *Carcharodon*, *Isurus*, *Lamna*, and related taxa where tooth material properties, wear, chipping, and replacement context can be observed directly.

Definition 7.3 (Ecological validation requirement)**.** An ecological interpretation is treated as validated only when the specimen-specific stress result is linked to at least one independent empirical closure condition: fossil damage consistency, extant analogue calibration, tooth-position invariance or quantified tooth-position variation, histological plausibility of the inferred fracture path, or fracture-mechanics feasibility under a documented tissue-property range. Without such closure, ecological interpretation remains hypothesis-generating.

Table 10: Reviewer-facing evidence upgrade checklist. The present DURF report contains a single-specimen tooth-level FEA; the items below describe what a palaeobiomechanics reviewer would reasonably request before the work is advanced into a formal species-level or ecology-linked manuscript.

Evidence layer	Present status	Required upgrade before stronger inference
Damaged megatooth specimen survey	Not claimed; only one physical fossil tooth is analysed	Curated multi-specimen dataset with damage class, location, preservation score, and specimen metadata
Tooth-position distribution	Not available in this report	Assignment of anterior, lateral, and posterior positions, plus separate damage and stress summaries by tooth position
Lamniform tooth histology	Cited only as comparative background	Histological constraints on enameloid, dentine, and EDJ architecture, preferably with extant and fossil comparison
EDJ tissue parameters	Treated as uncertain modelling inputs	Tissue-specific modulus, toughness, and interface-strength ranges; sensitivity analysis over EDJ parameters
Extant lamniform analogue	Deferred validation requirement	Matched comparison with *Carcharodon*, *Isurus*, *Lamna*, or related taxa under similar boundary conditions
Bite-mark, microwear, and chipping comparison	Not claimed as a dataset	Independent bite-mark or microwear comparison linking modelled high-stress regions to observed damage modes
Dental homology and tooth position	Outside present specimen-level claim	Systematic discussion of megatooth dental homology and position-specific functional loading
Sample-size expansion	Not part of this final report	Multi-tooth and ideally multi-locality sample across size class, preservation state, and geological context
Empirical validation dataset	Not yet available	Quantitative comparison between predicted high-stress masks and observed fossil damage distribution
Fracture-toughness constraint	Literature-based screening only	Realistic tissue-property interval from extant analogues, fossil preservation constraints, or mechanical testing where feasible

Table 11: Two manuscript routes should not be mixed without explicit evidence boundaries. The biomechanics-first route prioritises numerical verification; the palaeobiology route requires a biological comparison and fossil-damage anchor.

Submission route	Minimal defensible aim	Main evidentiary demand
Biomechanics-first manuscript	A reproducible μCT-to-FEA workflow showing serration-scale stress localisation in a fossil shark tooth under audited numerical assumptions.	FEA model definition, mesh convergence, boundary-condition sensitivity, material-parameter uncertainty, verification, validation, and reproducible data/code.
Palaeobiology/functional-morphology route	A functional-morphology hypothesis linking serration-scale stress localisation to fossil-damage predictions in megatooth sharks.	Tooth-damage distribution, tooth position, lamniform comparative context, fossil evidence, functional significance, and sample-size expansion.

7.7 Minimal Manuscript Upgrade Roadmap

The present report already has a defensible core: a specimen-derived tooth mesh, weak-form FEA setup, and region-wise stress-localisation audit. The next step is not additional narrative packaging but a validation loop that makes the model reproducible and externally checkable. The central claim for any manuscript version should remain:

> A new specimen-derived, single-mesh pressure calculation yields reproducible geometric-region percentiles. Its ordering is support-dependent; anatomical serration-tip amplification and fossil-damage predictions remain unverified hypotheses.

The claim should not be expanded into a complete feeding sequence, prey ceiling, *Livyatan* niche partition, maximum body size, whole-body weak point, ecology-superiority claim, or extinction mechanism.

7.7.1 Route A: biomechanics-first manuscript

A biomechanics-first manuscript should be built around a reproducible μCT-derived finite-element workflow. The required result set is: CT-to-mesh pipeline, segmentation sensitivity, mesh convergence, boundary-condition sensitivity, region-wise stress percentiles, material-uncertainty sweep, and fracture-screening discussion. The minimum figure/table set is: workflow schematic, CT/mesh reconstruction, curvature preservation, boundary/load conditions, mesh convergence, region-wise stress maps, sensitivity heatmap, CT metadata, mesh quality, material sweep, region-wise percentiles, and model limitations. Without mesh convergence and boundary-condition sensitivity, a biomechanics-first manuscript remains vulnerable.

7.7.2 Route B: palaeobiology / functional-morphology manuscript

A palaeobiology-style manuscript should be built around a functional-morphology hypothesis. The required result set is: tooth-level FEA, region-wise stress localisation, pilot fossil-damage comparison, extant lamniform literature comparison, tooth-position or dental-homology caution, and biological discussion. The minimum figure/table set is: specimen and anatomical orientation, μCT-to-mesh workflow, serration stress localisation, fossil-damage map or pilot survey, conceptual prediction diagram, evidence levels, specimen/image survey, FEA assumptions, and falsifiable predictions. Without a fossil-damage or comparative biological anchor, this route remains an engineering report rather than a palaeobiological argument.

7.7.3 Seven review-grade modules

1. **CT metadata audit.** Report $(\Delta x, \Delta y, \Delta z)$, grey-value unit, threshold protocol, threshold perturbation, and segmentation repeatability. The current spacing entry is $(50, 50, 100)$ μm, and grey values should remain scanner-native attenuation values unless calibrated HU is documented.

2. **Segmentation sensitivity.** Run low/baseline/high segmentations, for example baseline ±5%, and compare volume, surface area, serration notches preserved, curvature distributions, and stress-localisation mask overlap.

3. **Mesh convergence.** Use at least three mesh levels: coarse (~ 3 elements/notch), medium (~ 6–8 elements/notch), and fine (~ 12+ elements/notch). Report element count, DOF, element quality, $Q_{0.90}$, $Q_{0.95}$, $Q_{0.99}$, hotspot overlap, and peak stress only as a diagnostic. Preferred figures are $Q_{0.95}(R_i)$ versus mesh size h and Dice/hotspot-overlap versus h.

4. **Boundary-condition sensitivity.** Test root constraint, load direction, and load area. A minimal matrix is fully fixed / elastic foundation / partial fixed; normal / 30° oblique / lateral shear; narrow edge / crown band / broad labial patch. The target result is persistence or non-persistence of serration-band stress localisation.

5. **Material-model rationalisation.** Keep a linear-elastic baseline, one nonlinear sensitivity model, and a material uncertainty sweep in the main text. Move Ogden/MR2/MR5 details to supplement unless they are directly justified. A minimal sweep is E = 40–90 GPa, ν = 0.20–0.30, and pressure 0.5–5 MPa.

6. **Region-wise percentiles.** Table 9 should be script-reproducible from exported nodal or elemental stress fields. A minimum supplement should include `/data/stress_exports/`, `/masks/`, `/scripts/extract_region_quantiles.py`, `/tables/table9_region_quantiles.csv`, and a reproducibility README.

7. **Biological validation anchor.** Even a pilot survey is useful: the present physical tooth can be photographed and mapped, while public museum, auction, database, and literature images can provide a 20–50 specimen image-level pilot survey. The survey should record damage location as serration edge, crown face, root, or tip, and be labelled hypothesis-generating rather than population-conclusive.

7.7.4 Five-week minimum work plan

Week 1 Audit voxel spacing and grey-value units; fix coordinate system; perform segmentation threshold ±5%.

Week 2 Run three mesh levels and a small boundary-condition matrix; export stress fields.

Week 3 Define masks and automate Table 9 extraction; output CSV and plots.

Week 4 Build a 20–50 image pilot damage survey and simple location-frequency table.

Week 5 Choose Route A or Route B, move nonessential material to supplement, and prepare data/code availability.

Table 12: Final review-grade checklist for the next manuscript version.

Item	Required manuscript status
CT voxel spacing	Verified and reported as $(50, 50, 100)$ μm, with raw metadata retained.
Grey value / HU	Clarified as scanner-native attenuation unless HU calibration is documented.
Segmentation sensitivity	Completed with threshold perturbation and geometry/stress-overlap audit.
Mesh convergence	At least three mesh levels with percentile and hotspot-overlap reporting.
Boundary-condition sensitivity	At least three variants in root constraint, load orientation, or load area.
Material uncertainty	At minimum E/ν and load-pressure sweeps.
Table 9	Reproducible from scripts, masks, and exported stress fields.
Peak stress	Retained as diagnostic only, not as the main convergence or fracture result.
K_N	Treated as future metric unless annular fits, paths, mesh levels, and error estimates are provided.
Fossil damage anchor	At least a pilot image/specimen survey for palaeobiology route.
Whole-body ecology	Excluded from Results.
Data availability	Scripts, masks, stress exports, and README supplied.
Author claim	Kept aligned with actual single-specimen contribution.

7.8 Falsifiable Predictions for Future Work

The most useful consequence of the tooth-level model is not a dramatic whole-animal narrative but a set of testable predictions:

1. Damaged megatooth specimens should show non-random concentration of chips or spalls near serration-bearing cutting edges.
2. If EDJ deflection is important, damage should often follow the enameloid-dentine interface rather than form simple through-crown breaks.
3. Damage frequency should vary by tooth position because anterior, lateral, and posterior teeth experience different load orientations.
4. Extant lamniform comparison should show whether similar serration geometry produces similar local stress concentration under matched boundary conditions.
5. Dynamic contact models with prey-tissue compliance should test whether static stress-localisation results survive realistic oblique loading.

8 Conclusions

The revised manuscript separates direct computational outputs from broader functional speculation and defines the central claim narrowly.

1. The DICOM-to-FEA pipeline provides a reproducible method for turning a scanned anterior megatooth into a mechanics-ready tetrahedral model. The strongest evidence remains Level 1: topology, mesh quality, curvature preservation, and traceable solver outputs.

2. The new 1 MPa solve yields reproducible geometric-region percentiles. Its root-side band is highly stressed; a universal serration-tip-dominant ordering and a low-stress root control are not established.

3. The present high-resolution peak-stress table is retained for traceability but cannot yet be used as strong constitutive-model evidence, because identical maxima across all studies require raw COMSOL dataset audit.

4. Fracture-mechanics equations provide useful definitions for future testing, but a validated K_N, energy-release-rate, or cohesive-zone convergence study has not yet been demonstrated. The earlier numerical convergence claim is therefore downgraded to future work.

5. No complete-body mechanics, prey ecology, *Livyatan* comparison, maximum-size reconstruction, or Miocene ecosystem synthesis is claimed as a result of this manuscript.

6. The defensible numerical result is an explicitly specified linear-elastic pressure calculation on one cleaned volume mesh, with exact geometric masks, archived fields and equilibrium checks. Its regional ordering is conditional on the chosen supports and sampling measure.

7. Original Table 9 was not reproducible from the archive. Its numerical values and anatomical tip-dominant interpretation are superseded by the new proxy-region table. The A/B edge-to-crown $Q_{0.95}$ ratios are 2.024 and 1.521; this does not validate individual serration-tip stresses, mesh convergence, fracture, population-level damage frequency or ecological interactions.

Limitations and Future Work

1. **Single-specimen limitation.** One anterior tooth cannot represent the full morphological variation of *O. megalodon* across ontogeny, geography, tooth position, and geological time. The report therefore makes specimen-specific stress-localisation claims only. It does not claim a fossil damage-frequency dataset, a tooth-position distribution, a microwear dataset, or an extant lamniform calibration sample. A next step is a curated sample of anterior, lateral, and posterior teeth scanned and processed with the same pipeline.

2. **High-resolution data audit.** The identical reported peak stresses in the high-resolution table must be checked against raw COMSOL exports. A stronger version should report study-specific datasets, post-processing selections, load cases, and peak locations.

3. **Material-property uncertainty.** The current nonlinear constitutive models are tangent-matched screening models. A submission version should justify whether linear elasticity is sufficient, identify what the nonlinear parameters are fitted to, and propagate uncertainty in E, ν, K_{IC}, and EDJ toughness.

4. **Boundary-condition uncertainty.** Static or quasi-static pressure protocols are useful for localisation, but real feeding involves transient contact, oblique entry, prey compliance, tooth-row interaction, and head motion. Dynamic contact FEA should replace simplified loads before behavioural claims are strengthened.

5. **Mesh and fracture convergence.** The draft should report not only peak stress but also convergence of region-wise percentile stress, annular K_N fits, energy release rate, cohesive-zone damage, or finite process-zone averages. Peak von Mises stress alone is sensitive to local singularities.

6. **Comparative palaeobiological evidence.** The functional hypothesis should be tested against damaged megatooth specimens, tooth-position distributions, extant lamniform histology, tooth replacement mechanics, and serrated-tooth analogue models.

Table 13: Minimum additions needed before this DURF final technical report becomes a formal journal manuscript.

Issue	Current status	Required upgrade
Constitutive laws	Tangent-matched screening family	Material calibration or explicit linear-elastic justification
Peak stresses	Traceability record; identical high-res maxima need audit	Raw export table with model-specific peak location and dataset ID
Fracture metric	Screening equations only	K_N, G, or cohesive-zone convergence across mesh levels
Damage interpretation	Mechanically compatible hypothesis	Fossil damage survey and tooth-position comparison
EDJ pathway	Theoretical criterion only	Multi-material segmentation or extant lamniform EDJ parameters
Functional inference	Tooth-level only	Comparative validation before behavioural or ecological claims

Recommended next version. For a manuscript aimed at a palaeobiology or biomechanics venue, the most credible title remains narrow: *A μCT-to-FEA workflow for stress localisation in an anterior tooth of Otodus megalodon*. Any organism-level or ecological material should be placed in a separate supplementary model with explicit anatomical priors, uncertainty propagation, and independent validation.

Author Contributions and Supervision

Eliguli Han developed the computational framing of the study, assembled the μCT-to-mesh-to-FEA workflow, generated and organised the mechanical figures, performed the tooth-level palaeobiomechanical interpretation, and prepared the manuscript draft. Leslie Greengard is listed as faculty advisor for the project and is credited for supervisory guidance on the computational and mathematical framing. This version intentionally avoids presenting whole-organism reconstructions as results of the tooth FEA.

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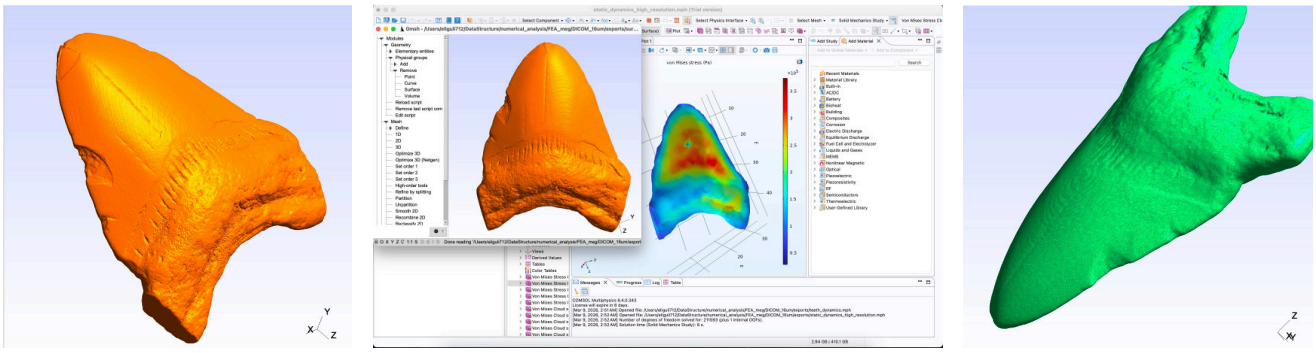
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A GUI Screenshots and Solver Session Records

The following appendix figure plates document the COMSOL and Gmsh GUI sessions from which the pipeline-rendered results in the main text are derived. These are included for reproducibility and workflow verification; the physically meaningful results are the pipeline-rendered figures in the main text.

Appendix figure style. The appendix plates are treated as provenance screenshots rather than publication figures. For a journal manuscript, these views should be redrawn or replaced by consistently scaled orthographic renders, shared colour bars, identical typography, and uniform panel labels. The current DURF report keeps them only to document the numerical workflow.

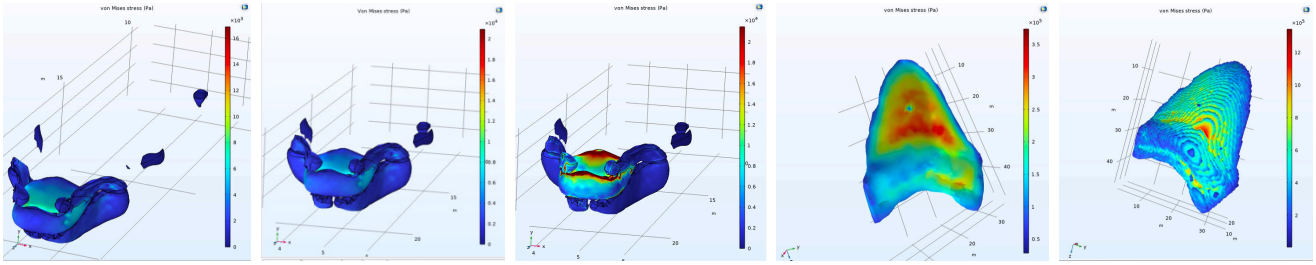
Excluded exploratory models. Jaw and whole-body screenshots from earlier exploratory work are not included as evidence in this version. They would require a separate comparative reconstruction protocol and are therefore excluded from



(a) DICOM surface, Gmsh. **(b)** Full-res DICOM, Gmsh. **(c)** Taubin-smoothed, Gmsh.

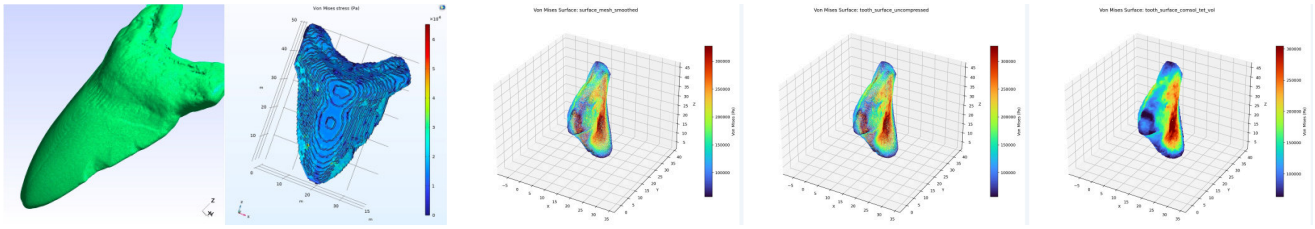
Figure 10: Gmsh GUI sessions: raw DICOM segmentation (a, b) and post-smoothing (c).

the main inferential chain.



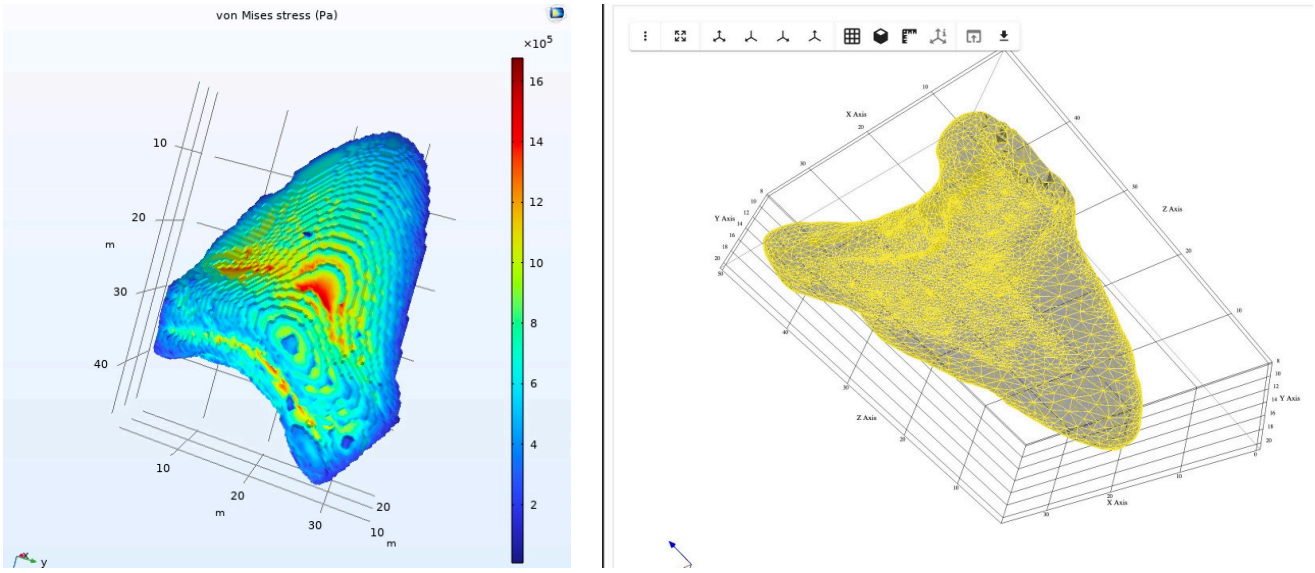
(a) Ogden (30k). **(b)** MR2 (30k). **(c)** MR5 (30k). **(d)** Ogden (343k). **(e)** NH (343k).

Figure 11: COMSOL GUI: low-res (a–c) and high-res (d–e) tooth FEA sessions. Von Mises Stress Cloud plots showing the solver model tree and recovery file history.



(a) Gmsh surface (left) + COMSOL σ_{vM} (right; 211,593 DOF). **(b)** σ_{vM} projected onto DICOM point cloud.

Figure 12: Pipeline bridge GUI views.



(a) Full-res mesh import (5.55M boundary elements). **(b)** Wireframe in stress_analysis.ipynb pipeline.

Figure 13: Mesh import and wireframe verification GUI views.